

Encoding an Engineered TNBC feature into a Qubit

Patrina Pun
Week 6 Undergrad

Why Do We Have To Turn A Classical Number Into a Qubit State?

Quantum Circuits don't take raw classical numbers :

- ❖ A quantum circuit operates on qubit states, the data should be turned on a rotation before any quantum gate can access it.
- ❖ Any numerical value for instance, 4, 6, 8 are classical values that must first be **normalized** and then be turned into a rotation angle, because quantum gates can only act on qubit states **not** on raw classical numbers directly.

For Today's Example:

Tumor size mm

- ❖ An Ideal Feature as it is a key measurement for doctors to distinguish cancer
 - ❖ A smaller tumors produce smaller rotations, while larger tumors produce larger rotations. This allows the quantum circuit to distinguish between different patients' tumor sizes.
- ❖ For instance, today we will utilize:

Tumor Size
25

Angle Encoding: is the bridge between the classical feature space and the quantum space state.

- 1) **Normalize**

$x_{\text{norm}} = \text{tumor_size_mm} - \text{min}) / (\text{max} - \text{min})$
- 2) **Angle Encode**

$\theta = \pi \cdot x_{\text{norm}}$
- 3) **Apply RY(θ)**

$\text{RY}(\theta) \mid \theta = \cos(\theta/2) \mid 0\rangle + \sin(\theta/2) \mid 1\rangle$

Example:

❖	tumor_size_mm	x_norm	$\theta = \pi \cdot x_{\text{norm}}$	$\cos(\theta/2)$	$\sin(\theta/2)$	Encoded State $\mid \psi \rangle$
❖	25	$(25-17)/23 = 8/23 = 0.3478$	$\pi \cdot 0.3478 = 62.6$	$\cos(62.6/2) = 0.8551$	$\sin(62.6/2) = 0.5184$	$\mid \psi \rangle = 0.855 \mid 0 \rangle + 0.518 \mid 1 \rangle$

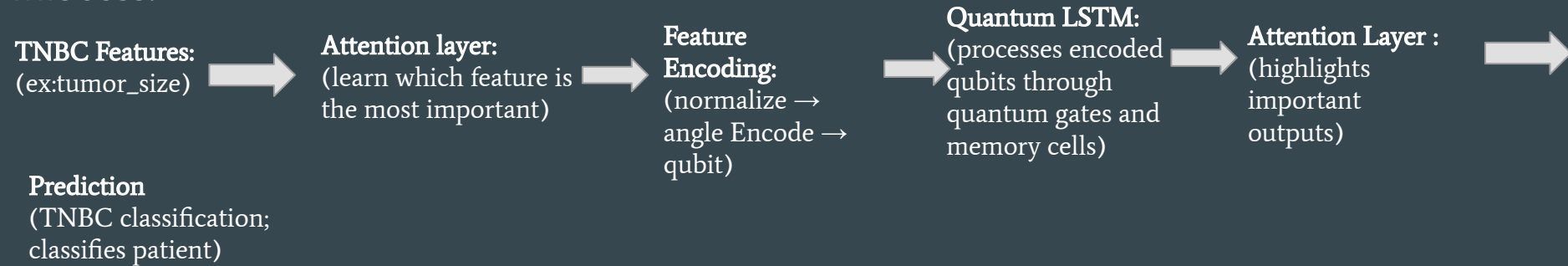
❖ tilted about **62.6°** away from $\mid 0 \rangle$ toward $\mid 1 \rangle$, sitting about one-third of the way down the Bloch sphere from the north pole.

Where Does Encoding Fit into A-QLSTM-A?

A-QLSTM-A: Attention Quantum Long Short-Term Memory Architecture:

- ❖ a hybrid machine learning model that combines attention mechanisms, quantum computing, and Long Short-Term Memory (LSTM) networks to analyze complex data and make predictions.

Process:



Continuation:

- ❖ Feature encoding is the bridge between classical TNBC data and the quantum portion of the A-QLSTM-A model. It converts engineered features into qubit states so the Quantum LSTM can process them.
- ❖ This is done by mapping feature values to qubit rotations, allowing the quantum circuit to process the information.

Closing:

- ❖ One engineered feature (tumor_size_mm) $\rightarrow$ normalized $\rightarrow$ turned into a rotation angle $\theta \rightarrow$ encoded into a qubit state $|\psi\rangle$.
- ❖ That qubit state is the handoff into the QLSTM - where the learning happens.
- ❖ Encoding builds the input; training/learning is what makes it predictive.